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May 30, 1973
SUPERSEDING
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November 3, 1964

FEDERAL SPECIFICATION

CUPS AND GLIDES, FURNITURE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cups, pads, shoes and glides placed under legs of furniture to protect floors and rugs.

1.2 Classification. Furniture cups, pads, shoes, and glides shall be of the following types and styles, as specified (see 6.2):

Type I - Cups, pads, and shoes.

Style 1 - Caster cups.
2 - Furniture pads.
3 - Desk shoes.

Type II - Glides.

Style 1 - Wood furniture legs.
2 - Metal furniture legs.

Type III - Special.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood-Cleated, Plywood.
PPP-B-621 - Boxes, Wood, Nailed & Lock Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government acticities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I. The type of rubber, rubber-like, or plastic material and the type of fabric pile shall be at the manufacturer's option provided they meet the performance requirements in paragraph 3.5.

3.2 Type I, cup, pads, and shoes.

3.2.1 Style 1, cups.

3.2.1.1 Shape. Cups shall be round or square, as specified (see 6.2). The interior, or furniture resting surface, may be ribbed or contoured and a caster rest may be provided at the manufacturer's option.

3.2.1.2 Base. The rug or floor bearing surface shall be resilient to prevent sliding on a hard smooth surface, contain extending prongs, or be covered by a fabric pile, as specified (see 6.2).

3.2.1.3 Size. Size of round and square cups shall be as follows, and as specified (see 6.2):

Round cups
Solid rubber and plastic
Inside diameter

Inches	Mm
1-1/4	32
1-3/8	35
1-5/8	41
1-3/4	41

Square cups
Inside dimensions
length and width

Inches	Mm
1-1/2	38
1-3/4	44
2	51
2-5/8	67

Fabric pile rug or floor surface
Outside diameter

Inches	Mm
2-1/2	63
3	76

3.2.2 Pads.

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3.2.2.1 Shape and surfaces. Pads shall be square and smooth on all visible surfaces. The rug or floor bearing surface shall be either resilient to prevent sliding on a hard smooth surface, contain extending prongs, or be adhesive coated with a removable paper backing, as specified (see 6.2).

3.2.2.2 Edge of pads. Pad shall have raised edges on 4 sides, 2 sides, or 1 side, as specified (see 6.2).

3.2.2.3 Size. Pads shall be 2, 3 or 3-3/4 inches (51, 76 or 95 mm) square, as specified (see 6.2).

3.2.3 Style 3, desk shoes.

3.2.3.1 Shape and surfaces. Desk shoes shall be square with a deep recess. All surfaces shall be smooth. Rug or floor bearing surface shall be elastic to prevent slipping.

3.2.3.2 Size. Recess of desk shoes shall accommodate the following desk leg sizes, as specified (see 6.2):

Inches	Mm.
1-1/4 to 1-3/8	32 to 35
1-1/2 to 1-5/8	38 to 41
1-3/4 to 1-7/8	44 to 78
2 to 2-1/8	51 to 54
2-1/4 to 2-3/8	57 to 60

3.2.4 Styles 1, 2, and 3.

3.2.4.1 Color. Unless otherwise specified (see 6.2), color of cups, pads, and shoes shall be brown or gray at the manufacturer's option. Value of gray, and value, hue, and chroma of the brown color shall be at the manufacturer's option. Color of fabric pile on cups shall be at the manufacturer's option.

3.2.4.2 Inside depth. Unless otherwise specified (see 6.2), inside depth of cups, pads, and shoes shall be at the manufacturer's option.

3.3 Type II glides.

3.3.1 Style 1, for wood furniture legs. Glides for wood furniture legs shall be any of the following designs, as specified (see 6.2).

3.3.1.1 Nail-on glides. Nail-on glides shall consist of (1) a steel nail embedded in rubber or a rubber-like material, encased in formed steel or molded plastic, or (2) a nail embedded in molded plastic with a fabric pile on the rug or floor bearing surface, as specified (see 6.2). The nail shall be nickel or brass (copper-zinc alloy) plated. Length of nail shall be 5/8 ± 1/8 inch (16 ± 3.2 mm). Base size, material, and coating shall be as follows and as specified (see 6.2):

Base diameter		Gliding surface
Inches	Mm	
7/8	22	Nickel plated steel
15/16	24	Nickel plated steel
15/16	24	Molded plastic
1	25	Fabric pile
1-1/8	29	Nickel plated steel
1-1/4	32	Nickel plated steel
1-1/4	32	Molded plastic
1-1/4	32	Fabric pile
1-1/2	38	Nickel plated steel

3.3.1.2 Swivel stem glides. Swivel stem glides shall fit holes in wood furniture of 0.210 to 0.213 inches (5.3 to 5.4 mm) diameter. Tilt shall be 26° ± 3°. Base of guides shall be 1-1/4 inches (32 mm) in diameter and shall be nickel plated.

3.3.1.3 Split stem glides. Split stem glides shall be suitable for insertion in 3/8 inch (9.52 mm) outside diameter (O.D.) sockets commonly used for casters. Base diameter shall be 1-1/4 or 1-1/2 inches (32 or 38 mm), as specified (see 6.2). Base shall be nickel plated formed steel. When specified (see 6.2), split stem glides shall fit 3/8 inch (9.5 mm) diameter holes, 1-5/8 inch (44 mm) in depth. Molded plastic base diameter shall be 1-3/16 inch (30 mm) diameter or zinc plated formed steel base diameter of 1-9/16 inch (40 mm), as specified (see 6.2).

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3.3.1.4 Formed spring steel stem. These glides shall have a molded plastic base. Stem shall fit a 3/8 inch (9.5 mm) O.D. socket commonly used for casters. Base diameter shall be 1-1/2, 2 or 2-1/2 inch (48, 51 or 63 mm), as specified (see 6.2).

3.3.1.5 Drive stem glides. Drive stem glides shall have a hollow tubular stem with a sharp edge for driving into end grain of wood. Base diameter shall be 1-1/2, 2 or 2-1/2 inches (48, 51 or 63 mm), as specified (see 6.2).

3.3.1.6 Glides with ferrules. Glides with ferrules shall be suitable for attaching to the end of legs by pushing in place.

3.3.2 Style 2 for metal furniture legs. Glides for metal furniture legs shall be any of the following designs, as specified (see 6.2).

3.3.2.1 Glides for square metal legs.

3.3.2.1.1 Press fit glides. Glides for square metal legs shall have a formed steel base, or plastic base, as specified (see 6.2). Glides shall have a press fit in the metal legs of sizes as specified (see 6.2). Cup size of glides shall be as follows and as specified (see 6.2):

Cup size

Inches	Mm	Inch	Mm
3/4	19	1-1/16	27
27/32	21	1-3/32	28
7/8	22	1-1/8	29
15/16	24	1-3/8	35
		1-1/2	38

3.3.2.1.2 Umbrella spring stem glides. Umbrella spring stem glides shall be designed for tapping in place using a hammer without injuring the rubber insert. Glides shall be rigid or tilting, as specified (see 6.2). Gliding surface shall be rubber, or nickel plated formed steel, as specified (see 6.2). Glides shall fit 7/8 inch (22 mm), 1-1/16 inch (27 mm), or any other square tube size, as specified (see 6.2).

3.3.2.2 Glides for round tubular legs.

3.3.2.2.1 Formed steel spring stem. These glides shall have a U-shape formed steel spring stem. Base of glides shall be nickel plated. Unless otherwise specified (see 6.2), glides shall have a 1-1/2 inch (38 mm) diameter base. Stems for fitting the applicable tube size shall be as follows and as specified (see 6.2):

Tube diameters

Outside		Inside	
Inches	Mm	Inches	Mm
1-1/2	38	1.370	34.8
1-1/2	38	1.402	35.6

3.3.2.2.2 Umbrella spring stem glides. Umbrella spring stem glides shall be designed for tapping in place using a hammer, without injuring the rubber insert. Gliding surface shall be nickel plated formed steel. Base diameter and applicable tubular leg sizes shall be as follows, and glides shall be rigid or tilting, as specified (see 6.2):

Glides	Base diameter		Tube diameter			
			Outside		Inside	
	Inches	Mm	Inch	Mm	Inch	Mm
Rigid	15/16	24	3/4	19	0.625	15.9
Rigid	1-1/8	29	1	25	.870	22.1
Rigid	1-1/4	32	1	25	.870	22.1
Tilting	1-1/4	32	7/8	22	.777	19.7
Tilting	1-1/4	32	1	25	.870	22.1

3.3.2.2.3 Rubber expanding stem. The stem shall be rubber held in place by a flat metal nut on the central threaded stem. Dimensions of the complete stem assembly including thread size shall be at the manufacturer s option. Base diameter and applicable tubular leg size shall be as follows and as specified (see 6.2):

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Base diameter		Tube diameter			
		Outside		Inside	
Inches	Mm	Inch	Mm	Inch	Mm
15/16	24	3/4	19	0.527	13.4
15/16	24	7/8	22	.777	19.7
1-1/8	29	1	25	.902	22.9
1-1/2	38	7/8	22	.777	19.7
1-1/2	38	1	25	.828	19.0

3.3.2.3 Glides for square and round tubular legs.

3.3.2.3.1 Swivel glides. Swivel glides shall consist of a ball and socket rubber cushion. Base diameter shall be 1-1/8 or 1-7/32 inches (29 or 3 mm). Stem shall be umbrella design for inserting into metal legs. Glides shall swivel 22-1/2 ± 3 degrees. Glides shall be designed for round or square tube furniture legs and for the following tube sizes, as specified (see 6.2):

Round tube and square tube Outside dimensions			
Inches	Mm	Inches	Mm
5/8	16	1	25
3/4	19	1-1/16	27
13/16	21	1-1/8	29
7/8	22	1-1/4	32

3.3.2.3.2 Threaded stem. Threaded stem glides shall be as specified, including stem size, base diameter, and glide surface (see 6.2).

3.3.2.3.3 Grip stem. Grip stem slides shall have a stem with integral annular bands at ends and middle of stem. Diameter of stem, glide base diameter, and surface shall be as specified (see 6.2).

3.3.2.3.4 Plain stem. These glides shall have a plain stem. Diameter of stem, glide base diameter, and surface shall be as specified (see 6.2).

3.3.3 Styles 1 and 2.

3.3.3.1 Electroplated steels. Unless otherwise specified (see 6.2), thickness or weight of plating shall be at the manufacturer s option.

3.3.3.2 Dimensional tolerances. Unless otherwise specified (see 6.2), dimensional tolerances shall be at the manufacturer s option.

3.4 Type III, special. This type of cup, pad, shoe, and glide is defined as any type which does not fall within the other type classifications of this specification, but the materials, workmanship, and quality shall meet the applicable provisions of this specification.

3.5 Performance. Cups, pads, shoes, and glides shall not fracture nor be permanently deformed, and shall not mark, chemically react with, or color migrate to the surface upon which they are tested (see 4.4.2 and 4.4.3).

3.6 Marking. Cups and glides shall be permanently marked with the manufacturers name or trade mark.

3.7 Workmanship. Cups shall be free from chips, sharp edges and rough areas that affect function and serviceability. Glides shall be free from sharp edges. The elastomer part of glides shall not be cracked or split. Metals shall be free from cracks. Coating on metals shall cover all exposed surfaces except edges. Fabric pile shall be free from bare areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling. The sample size levels and the accpetable quality level (AQL s) shall be in accordance with MIL-STD-105.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of each type and style of cups and glides submitted for inspection at one time.

4.2.2 Sampling for dimensional and visual inspection of end item. Sampling of cups and glides for these characteristics shall be in accordance with level S-3.

4.2.3 Sampling for tests. Samples for test shall be in accordance with level S-2.

4.3 Examination and tests.

4.3.1 End item. Cups and glides selected in 4.2.2 shall be examined in accordance with table I. The AQL's for defects listed in table I shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Type and style not as specified.	X	
Edges of cups chipped or sharp.	X	
Surface of cups not smooth.		X
Size of cups not as specified.	X	
Color of cup not as specified when required.	X	
Inside depth of cups not as specified when required.	X	
Glide design not as specified.	X	
Size and gliding surface of glides not as specified.	X	
Sharp edges on glides.	X	
Elastomers in glides cracked or split.	X	
Cracks in metal parts of glides.	X	
Bare areas on fabric pile glides.	X	
Marking missing on cups and glides.		X
Damage or defects affecting function or serviceability.	X	
Damage or defects not affecting function or serviceability.		X

4.3.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance with table II. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operation. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers fully prepared for delivery. The lot size shall be number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container, loose strapping, inadequate stapling. Bulging or distortion of container.

4.3.3 AQL tests. The AQL for tests shall be 4.0 percent defective.

4.4 Test methods.

4.4.1 Deformation. Place cups and glides on a horizontal surface of unpainted plywood and apply a vertical load of 500 pounds (227 kg) for 30 minutes to the center of the cup or glide over an area of 3/4 inch (19mm) in diameter. After 30 seconds release the load and examine cup, glide and plywood (see 3.5).

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4.4.2 Color migration. Place cups and glides having plastic or fabric pile on a piece of asphalt or vinyl asbestos tile on a flat surface and apply a load as in 4.4.1 to the central area of cups and glides for a period of 24 hours. Release the load and examine the contacting surfaces (see 3.5).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Four cups or glides four cups comprising 1 set shall be packaged together in a bag fabricated of a transparent plastic material not less than 0.0020 (2 mils, 0.0508 mm) thickness or in a close-fitting box conforming to PPP-B-566, PPP-B-665 or PPP-B-676. The box shall be closed in accordance with the appendix to the applicable box specification.

5.1.1.2 Intermediate packaging. Ten unit packages of cups or glides shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665 or PPP-B-676. The box shall be closed in accordance with the appendix to the applicable box specification.

5.1.2 Level B. Cups and glides shall be packaged as in 5.1.1 and 5.1.1.2.

5.1.3 Level C. Cups and glides shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Cups and glides shall be packed in close-fitting boxes conforming to PPP-B-585, class 3, PPP-B-591, class II, PPP-B-601, type III, PPP-B-621, class 2, PPP-B-636, V3c, or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed approximately 200 pounds (91 Kg). The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Ten intermediate packages shall be packed in a close-fitting box conforming to PPP-B-636, class-domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. Cup and glides shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with National Motor Freight Classification or Uniform Freight Classification.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 of MIL-STD-129, as applicable (see 6.2).

6. NOTES

6.1 Intended use. Cups and glides are intended to be used under furniture legs for the protection of floors and floor covering against marks and scratches. Cups and glides with a soft elastomeric contact with floors and having a high coefficient of friction are also intended to prevent laterial movement of furniture.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

Cups and glides.

(a) Type and style (see 1.2).
(b) Size of lot if different from 4.2.1.
(c) Level of packaging and packing required (see 5.1 and 5.2).

Styles 1, 2, and 3, cups, pads, and shoes.

(a) Shape, base, and size of cups (see 3.2.1.1, 3.2.1.2 and 3.2.1.3).
(b) Base, edge, and size of pads (see 3.2.2.1, 3.2.2.2, and 3.2.2.3).
(c) Size of desk shoes (see 3.2.3.2).
(d) Color, and inside depth of cups, pads, and shoes, when required (see 3.2.4.1 and 3.2.4.2).

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Style 1, glides
(a) Design (see 3.3.1).
(b) Nail-on glides, design and size (see 3.3.1.1).
(c) Base diameter of formed spring steel and drive stem glides (see 3.3.1.4 and 3.3.1.5).
Style 2, glides
(a) Design (see 3.3.2).
Square tube furniture legs.
(a) Press fit glides (see 3.3.2.1.1).
(b) Umbrella spring stem glides (see 3.3.2.1.2).
Round tube furniture legs.
(a) Base and stem size (see 3.3.2.2.1).
(b) Base size and rigidity, umbrella spring stem glides (see 3.3.2.2.2).
(c) Base and leg size, rubber expanding stem (see 3.3.2.2.3).
Square and round furniture legs.
(a) Swivel glides, leg shape and size (see 3.3.2.3.1).
(b) Threaded stem glides (see 3.3.2.3.3 and 3.3.2.3.2).
Style 1 and 2
(a) Thickness or weight of electroplated glide surface when required (see 3.3.3.2).
(b) Dimensional tolerances, when required (see 3.3.3.3).

6.3 Caster cups and furniture pads. Caster cups have a flat, slightly curved, or indentation on the leg bearing surface, and pads have a uniform flat or ribbed surface on the leg bearing side and raised edges on one, two or all four sides.

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activity:

GSA-FSS

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